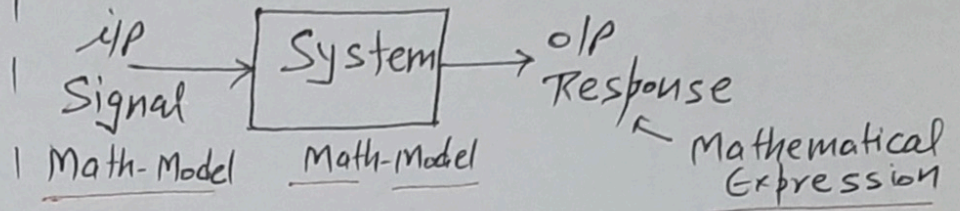


Syllabus of Signal & System (SAS)

1. Signals & classification
2. standard signals or
Basic signals or singular func.
3. Transformation in independent
Variable of signal or operation
of signals.
4. System & its Type
5. linear-Time invariant system
System Response using
Convolution method
6. Fourier Series
7. Fourier Transform
8. Laplace Transform
9. Z-Transform
10. Sampling of signal



Importance of "SAS"

* University Exam ECE, EEE, Instan.

* SAS $\rightarrow$ Communication System

$\Rightarrow$ Competitive Exam

ESE $\rightarrow$ objective + Subjective
(20-25 ques) ($\approx$ 40 Marks)

Gate $\rightarrow$ 10 to 14 Marks
Every year

PSU $\rightarrow$ key subject

Base of Communication System

Follow "SAS" lecture Series

Thank you.

Signals

Def:- Any entity having associated information with it is called as signal

* Electrical information are Present in form of voltage and current both function of time.

$$\left. \begin{matrix} f(t) = V(t) \\ I(t) \end{matrix} \right\} \text{input signal.}$$

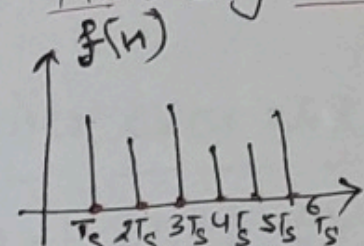
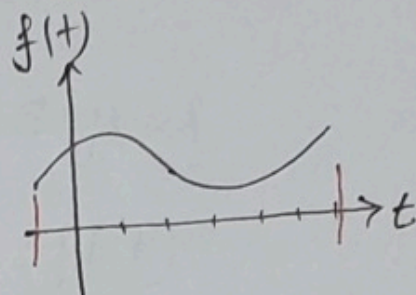
Classification of Signals

① Real Valued and Complex Valued Signals

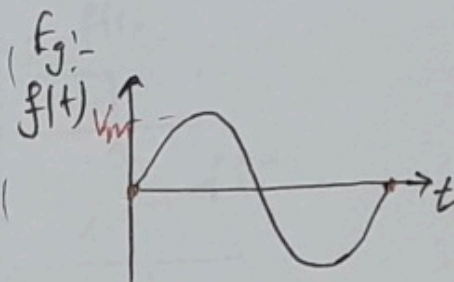
$$\begin{matrix} f(t) = st \\ t \rightarrow 0-\infty \\ f(t) \rightarrow \text{Real} \end{matrix}$$

$$\begin{matrix} f(t) = \frac{st}{R} + \frac{j7}{\text{Ime.}} \\ f(t) = j7 \end{matrix}$$

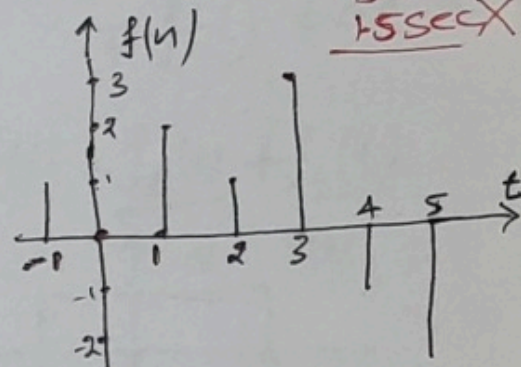
② continuous time & Discrete time signal.



$T_s = 1 \text{ sec}$
 $1.5 \text{ sec} \times$



$$f(t) = V_m \sin \omega t$$



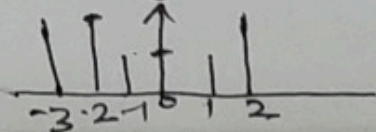
$$f(n) = \begin{cases} 1, 0, 2, 1, 3, -1, -2 \\ (-1) \uparrow (1) (2) (3) (4) (5) \end{cases}$$

origin

Arrow shows value of origin

Left side of $\uparrow \rightarrow$ -ve time

Right " " $\uparrow \rightarrow$ +ve time



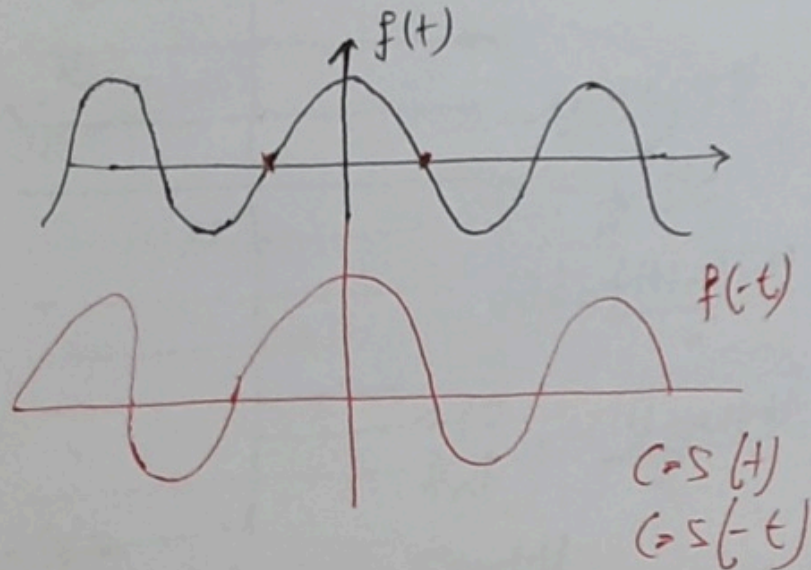
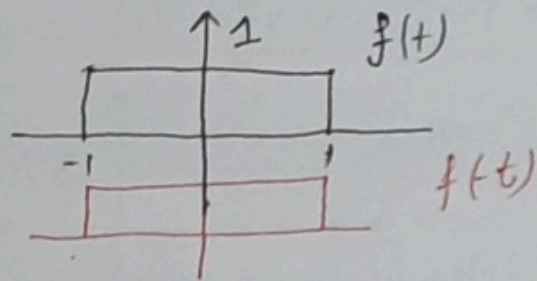
$$f(n) = \begin{cases} 2, 2, 1, 0, 1, 2 \\ \uparrow \end{cases}$$

Symmetric & non-Symmetric Signals

Even & odd signals.

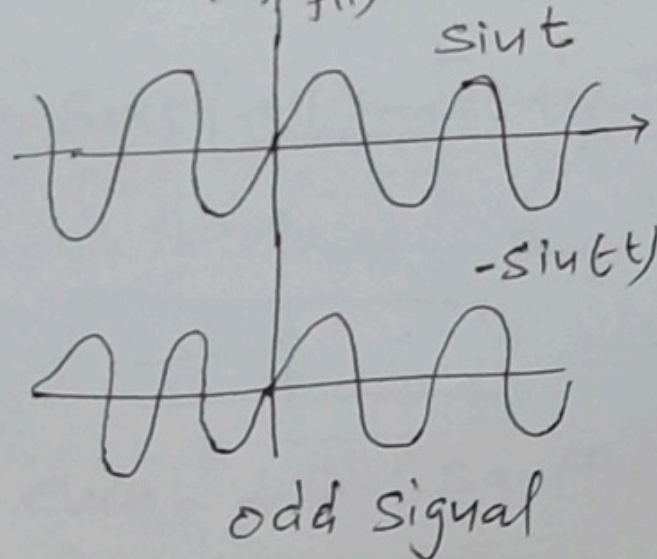
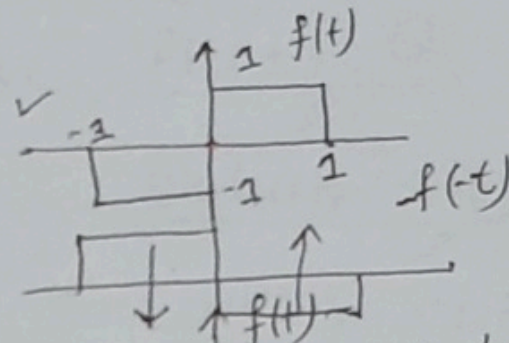
Even Signal

$$f(t) = f(-t)$$



Odd Signal

$$f(t) = -f(-t)$$

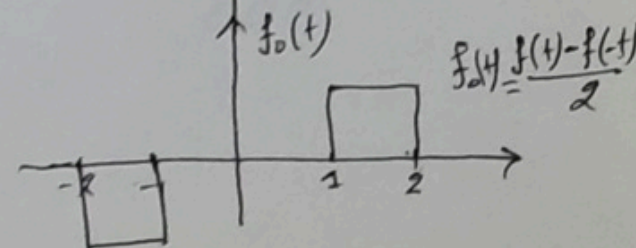
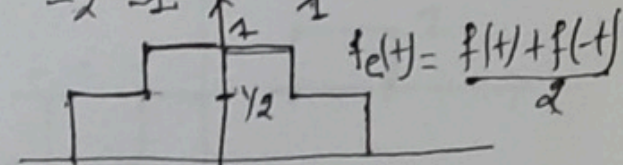
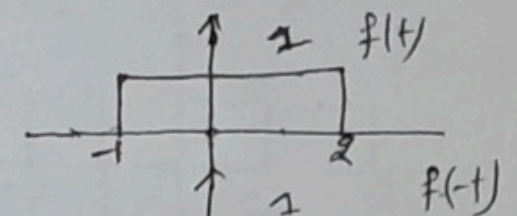


Neither even nor odd

$$f(t) \neq f(-t) \text{ even X}$$

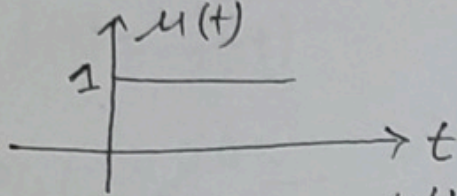
$$f(t) \neq -f(-t) \text{ odd X}$$

$$f(t) = f_e(t) + f_o(t)$$



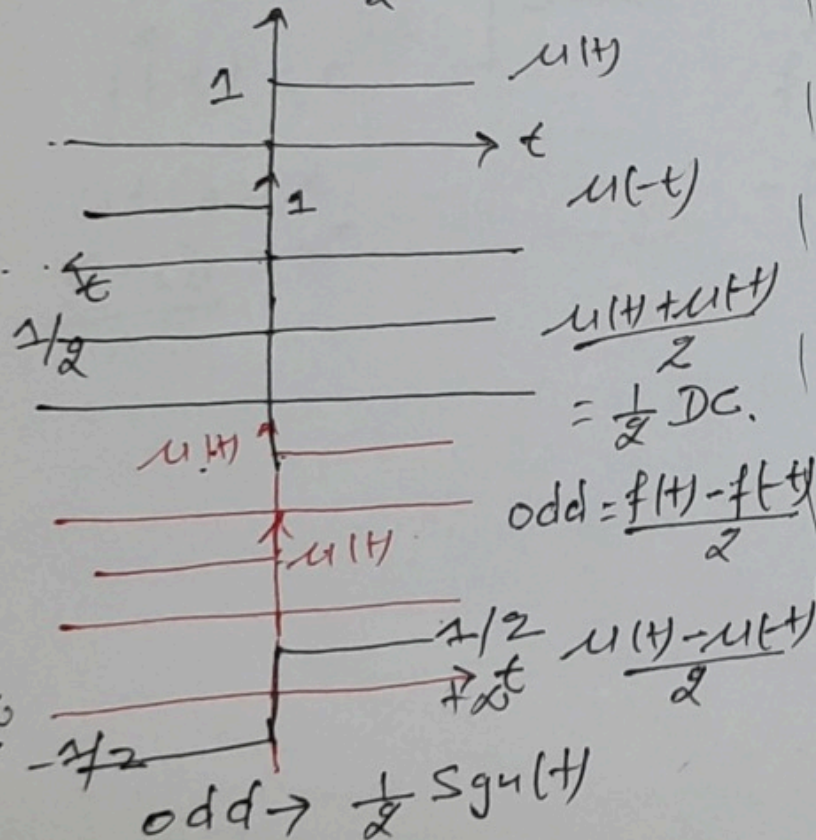
Problems of Even & odd signals Real valued s/g

① find even & odd Parts of given signal

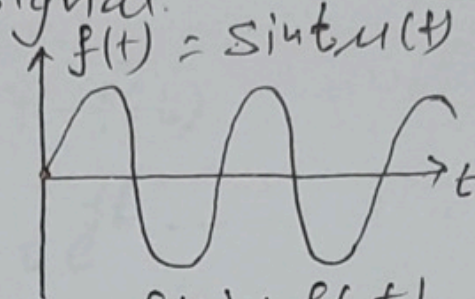


even = $\frac{f(t) + f(-t)}{2}$

$u(t) \rightarrow$ even
 $\rightarrow$ odd



② Find Even Parts of Signal.

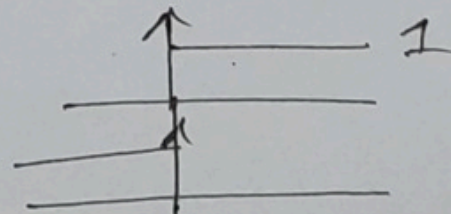


even = $\frac{f(t) + f(-t)}{2}$

$= \frac{\sin t u(t) + \sin(-t) u(-t)}{2}$

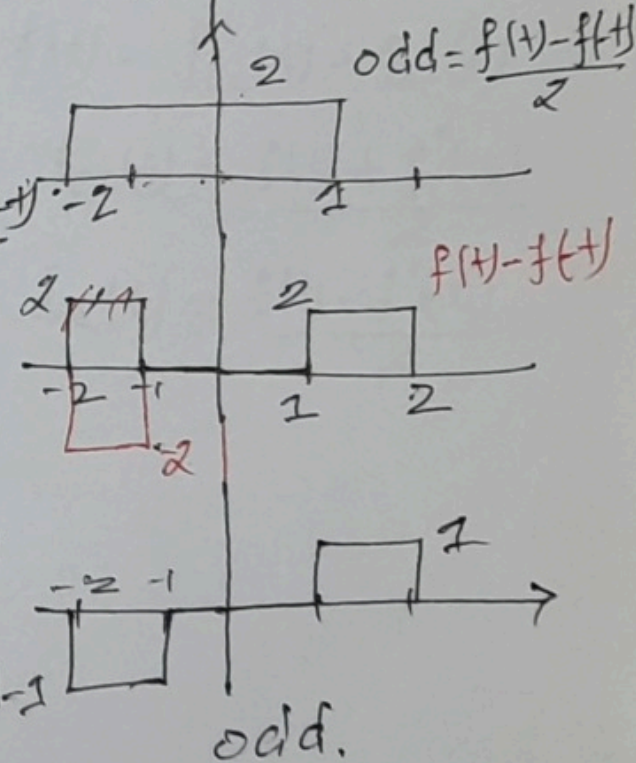
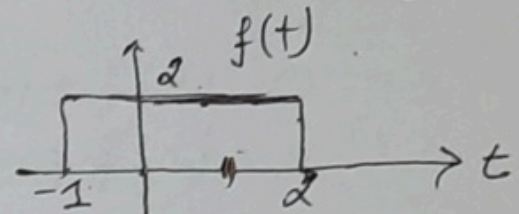
$= \frac{\sin t u(t) - \sin t u(-t)}{2}$

$\sin t \left(\frac{u(t) - u(-t)}{2} \right)$



even = $\frac{1}{2} \sin t \text{sgn}(t)$

③ Sketch the odd Parts of this signal



Even & odd for Complex signals

Even conjugate

$$f(t) = f^*(-t)$$

$$\left. \begin{aligned} f(t) &= e^{jt} \\ f(-t) &= e^{-jt} \\ f^*(-t) &= e^{jt} \end{aligned} \right\} \text{Same}$$

$$\begin{aligned} a+ib &= x \\ \underline{a-ib} &= x^* \quad \checkmark \end{aligned}$$

Odd conjugate

$$f(t) = -f^*(-t)$$

$$\left. \begin{aligned} f(t) &= te^{jt} \\ f(-t) &= -te^{-jt} \\ f^*(-t) &= -te^{jt} \\ -f^*(-t) &= te^{jt} \end{aligned} \right\} \checkmark$$

neither even nor odd conjugate

$$\begin{aligned} f(t) &\neq f^*(-t) \quad \text{even X} \\ &\neq f^*(t) \quad \text{odd X} \end{aligned}$$

$$f(t) = f_{ec}(t) + f_{oc}(t)$$

$$f_{ec}(t) = \frac{f(t) + f^*(-t)}{2}$$

$$f_{oc}(t) = \frac{f(t) - f^*(-t)}{2}$$

Problems of Even & odd of complex signal.

① $f(n) = \begin{bmatrix} -10+6j & 8-4j & -1+2j & -4-8j \end{bmatrix}$ find $f_{oc} = \frac{f(n) - f^*(n)}{2}$

$$f(-n) = \begin{bmatrix} 0 & -4-8j & -1+2j & 8-4j & -10+6j \end{bmatrix}$$

$$f^*(-n) = \begin{bmatrix} 0 & -4+8j & -1-2j & 8+4j & -10-6j \end{bmatrix}$$

$$f(n) - f^*(n) = \begin{bmatrix} -10+6j & 12-12j & 4j & -12-12j & 10+6j \end{bmatrix}$$

$$\frac{f(n) - f^*(n)}{2} = \begin{bmatrix} -5+3j & 6-6j & 2j & -6-6j & 5+3j \end{bmatrix}$$

Notes

odd $\times$ odd = Even sig

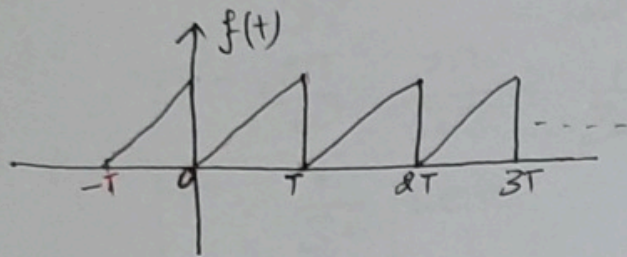
odd $\times$ even = odd sig

Even $\times$ odd = odd sig

— $\times$ even = even sig

Continuous Time Periodic & Aperiodic signals

Periodic Signal

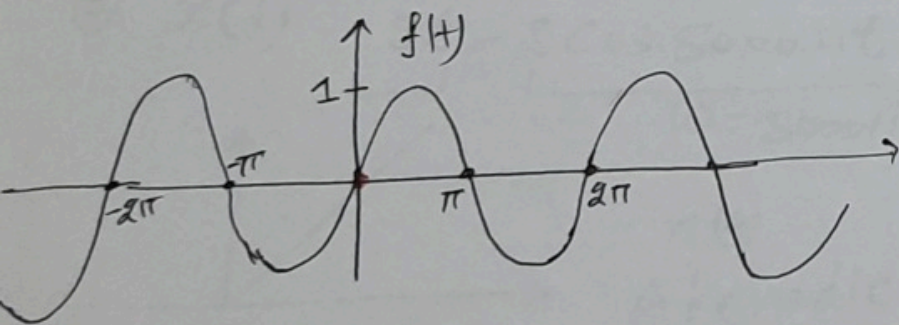


$T \rightarrow$ fundamental Period

$KT \rightarrow$ Time Period of s/g

$$f(t) = f(t \pm KT)$$

$\hookrightarrow$ Periodic

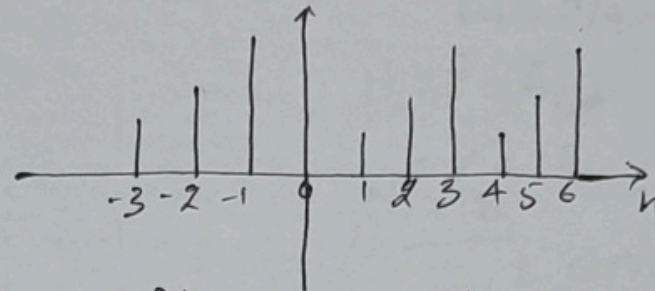


Time Period = 2π

$$x(t) = \sin \omega t \rightarrow T = \frac{2\pi}{\omega}$$

T can be any Real Value.

Discrete Time Periodic Signal



After 3 sample, value Repeat itself.

$$e^{j\omega n} \rightarrow \text{Time Period}$$

$$N = 8 \times \frac{2\pi}{\omega}$$

$\downarrow$ integer (smallest)
 $\downarrow$ Integer

$$e^{j3\pi n} \rightarrow N = 8 \times \frac{2\pi}{\omega}$$

$$\omega = 3\pi$$

$$= 8 \times \frac{2\pi}{3\pi}$$

$$= 8 \times \frac{2}{3} = 3 \times \frac{2}{3}$$

$$N = 2 = 2$$

$$\omega = \sqrt{2}\pi$$

$$e^{j\sqrt{2}\pi n} \rightarrow N = 8 \times \frac{2\pi}{\omega}$$

$$N = 8 \times \frac{2\pi}{\sqrt{2}\pi}$$

$$N = 8 \times \frac{2}{\sqrt{2}}$$

$\hookrightarrow$ never integer

$\hookrightarrow$ not Periodic Signal. X

Problems to find Period of Continuous & Discrete s/g

① $x(t) = \underbrace{14}_{DC} + \underbrace{50 \cos 60\pi t}_{AC}$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{60\pi} = \frac{1}{30}$$

$$T = \frac{1}{30} \text{ sec}$$

② $x(t) = 10 \sin 5t - 4 \cos 7t$

$$\omega_1 = 5 \quad \omega_2 = 7$$

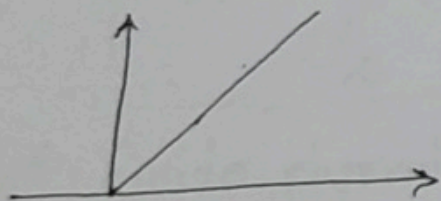
$$T_1 = \frac{2\pi}{5}, \quad T_2 = \frac{2\pi}{7}$$

$$LCM(T_1, T_2) \Rightarrow LCM\left(\frac{2\pi}{5}, \frac{2\pi}{7}\right)$$

$$\frac{2\pi}{1} \rightarrow T = 2\pi$$

③ $x(t) = \underbrace{5t}_{DC} - \underbrace{2 \cos 5000\pi t}_{AC}$

$$\omega = 5000\pi$$



Aperiodic

T X

$$\begin{array}{r} 18-14 \\ 4-7 \\ 18 \\ \times 7 \\ \hline 126 \end{array}$$

④ $x(n) = \cos \frac{\pi n}{4} + \sin \left(\frac{\pi n}{7} + \frac{1}{2} \right)$

$$\omega_1 = \pi/4 \quad \omega_2 = \pi/7$$

$$N_1 = 8 \times \frac{2\pi}{\pi/4} \quad N_2 = 8 \times \frac{2\pi}{\pi/7}$$

$$= 8 \times 18$$

$$N_1 = 18$$

$$N_2 = 14$$

$$N_1 = 18$$

$$LCM(18, 14) = 126$$

⑤ $x(n) = \cos\left(\frac{\pi n}{2}\right) - \sin\left(\frac{\pi n}{8}\right) + 3 \cos\left(\frac{\pi n}{4} + \frac{\pi}{3}\right)$

$$N_1 = 8 \times \frac{2\pi \times 2}{\pi} = 4$$

$$N_2 = 8 \times \frac{2\pi \times 8}{\pi} = 16$$

$$N_3 = 8 \times \frac{2\pi \times 4}{\pi} = 8$$

$$LCM(4, 16, 8) \rightarrow 16$$

⑥ $x(n) = 2 e^{j\left(\frac{n}{6} - \pi\right)}$

$$x(n) = 2 e^{j\pi} e^{jn/6} \quad \omega = 1/6$$

$$N = 8 \times \frac{2\pi}{1/6}$$

$$= 8 \times 12\pi$$

↳ not possible

Periodic X

4	4-16-8
2	1-4-2
	1-2-1

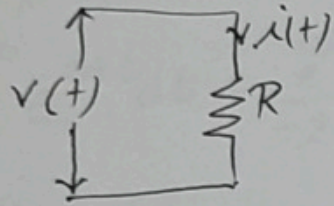
⑦

Energy & Power signal

Energy signal

* Finite Energy

* Zero Power



$$P(t) = \frac{v^2(t)}{R} = v^2(t) \quad \left\{ \begin{array}{l} \\ f^2(t) \end{array} \right.$$

$$= i^2(t) \cdot R = i^2(t)$$

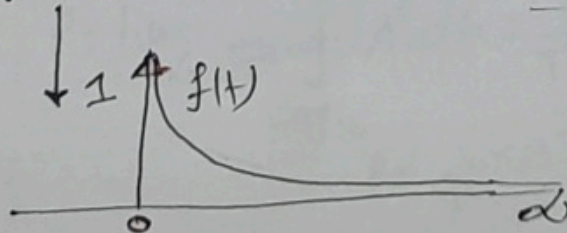
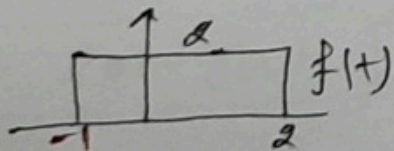
if $R=1 \rightarrow$ Normalisation

$$P(t) = \frac{dE}{dt} \Rightarrow E(t) = \int_{-\infty}^{\infty} P(t) \cdot dt$$

$$E(t) = \int_{-\infty}^{\infty} |f^2(t)| \cdot dt \quad \text{xxx}$$

* finite duration & finite Amplitude

* infinite duration but value tends to zero as $t \rightarrow \infty$



Power signal

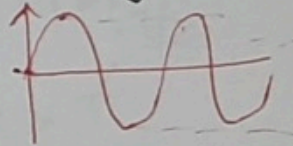
* finite Power but ∞ Energy

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt \quad f(t) \rightarrow |f(t)|$$

* Power signal infinite duration

* $t \rightarrow \infty$ value tends to constant

* Periodic signals are generally Power signal.

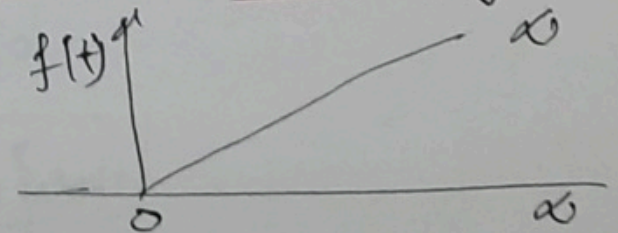


* if for a signal $f(t)$

$$\text{Power} = \infty$$

$$\text{Energy} = \infty$$

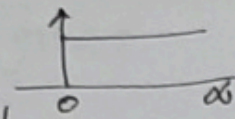
Then This signal is neither Power nor Energy signal.



Problems of Energy & Power Signals

① $f(t) = e^{-at} u(t)$

find E of $f(t)$



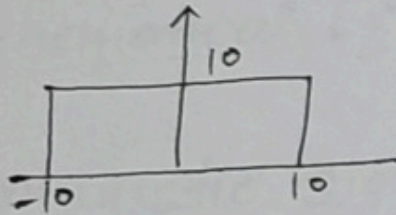
$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

$$E = \frac{1}{2a} \text{ Joule}$$

$$E = \int_{-\infty}^{\infty} (e^{-at})^2 dt$$

$$E = \int_0^{\infty} e^{-2at} dt = \left. \frac{e^{-2at}}{-2a} \right|_0^{\infty} = 0 - \left(\frac{1}{-2a} \right)$$

② $f(t)$



find Energy of $f(t)$

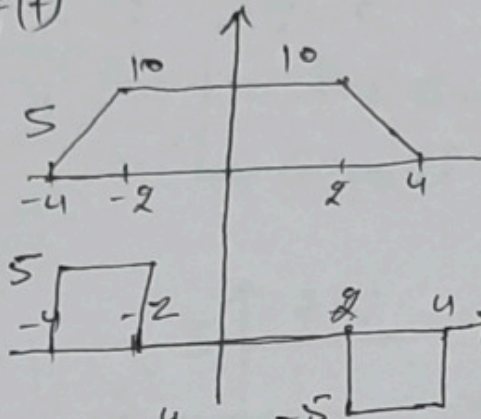
$$E = \int_{-10}^{10} (10)^2 dt = \int_{-10}^{10} 100 dt$$

$$E = 100 \left[t \right]_{-10}^{10}$$

$$= 100(10 - (-10))$$

$$E = 2000 \text{ Joule}$$

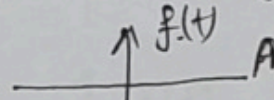
③ find E of derivative of $f(t)$



$$E = \int_{-4}^{-2} 5^2 dt + \int_{-2}^2 5^2 dt + \int_2^4 5^2 dt$$

$$E = 25 \left[t \right]_{-4}^{-2} + 25 \left[t \right]_{-2}^2 + 25 \left[t \right]_2^4 = 50 + 50 = 100 \text{ Joule}$$

④ Find Power of given Signal



$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} A^2 dt = \frac{1}{T} \times A^2 \left[t \right]_{-T/2}^{T/2}$$

$$P = \frac{1}{T} \times A^2 \times T = A^2 \text{ Watt}$$

⑤ $y(2t + \frac{\pi}{4})$
 $x(t) = e^{j\pi/4} e^{j2t}$

Power or Energy Signal

$$x(t) = e^{j\pi/4} e^{j2t} \quad \text{N.T.E} \rightarrow |e^{j2t}| = 1$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \times 1 \times T$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \times 1 \times T$$

$$P = 1 \text{ Watt}$$

Random & Deterministic Signal.

Random Signal Noise

- * future value of signal not known at Present time.
- * can not be Presented in Mathematical Eqⁿ or Expression.

Deterministic Signal

- * future value of signal is known at Present time
- * Represented by well defined mathematical Expression.

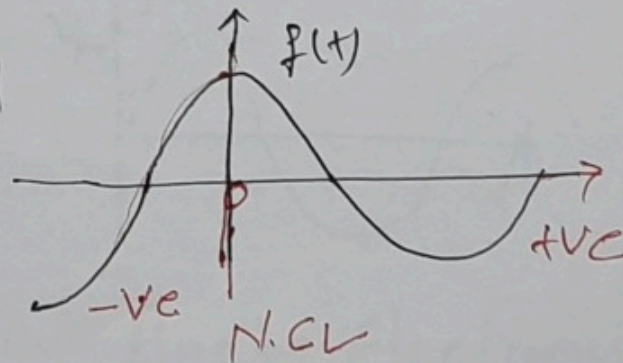
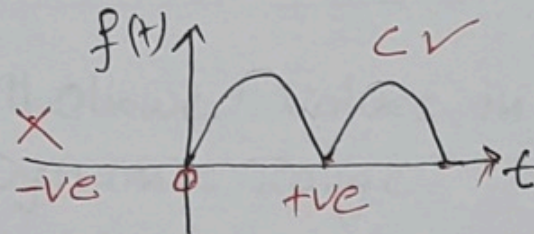
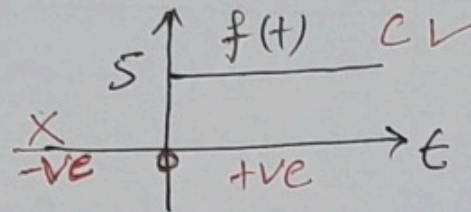
$$\underline{f(t)} = 2x(t) + 5$$

$t = 5 \text{ Sec}$

Causal & Non-Causal Signal

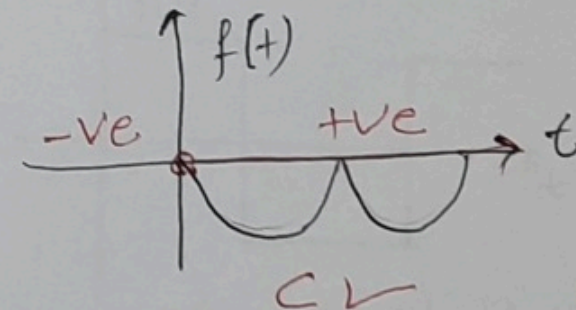
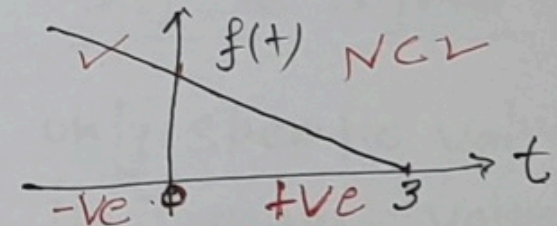
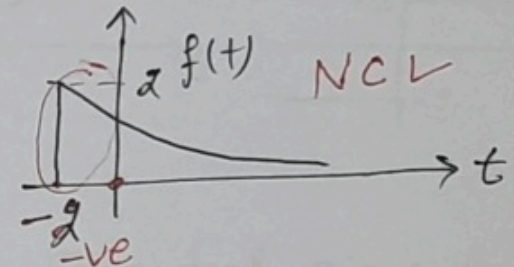
if $f(t) = 0$

for $t < 0$
→ Causal ✓



if $f(t) \neq 0$

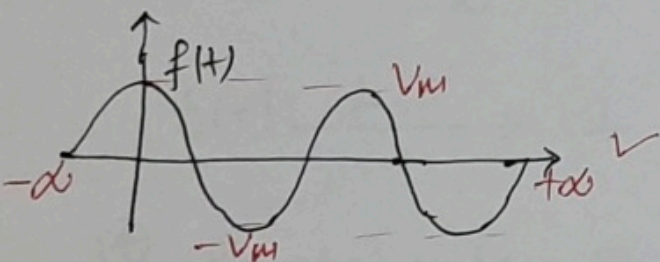
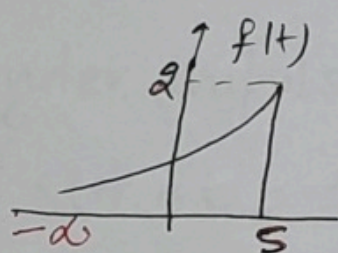
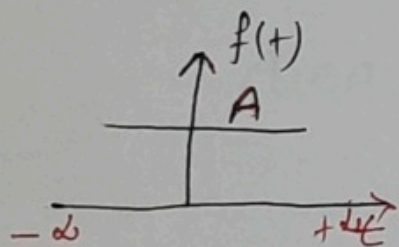
for $t < 0$
→ Non-Causal



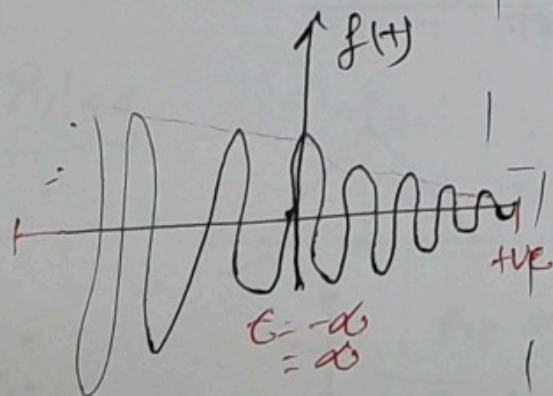
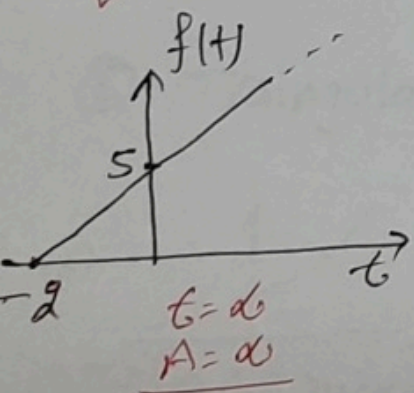
Bounded & unBounded signal

* Bounded signal

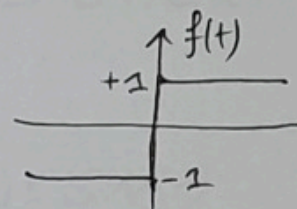
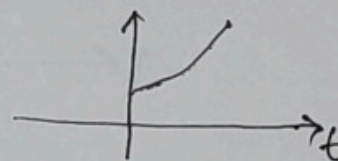
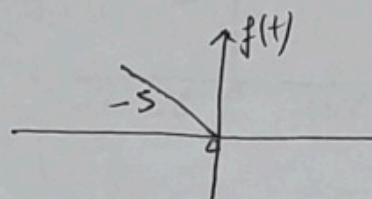
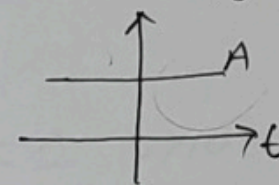
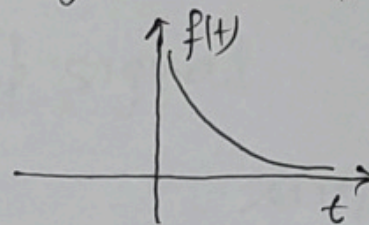
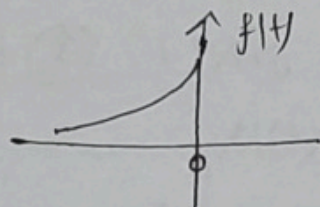
$f(t) \leq N \rightarrow$ finite for all t



unbounded signal ✓

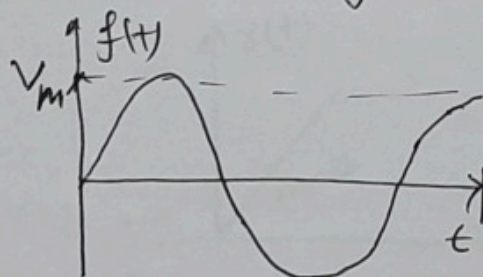


Left Sided, Right sided, Two Sided s/g



Analog Signal

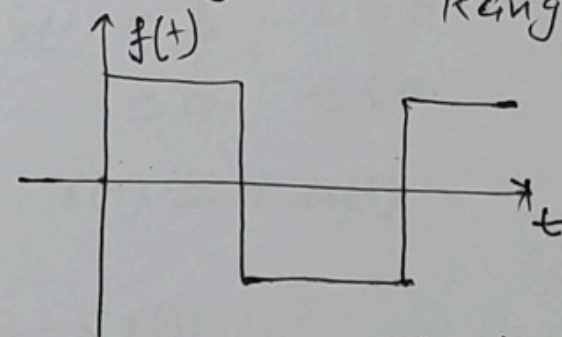
All allowed values in Dynamic Range



Analog Electronics input signals

Digital Signal

only specific value in Dynamic Value Range

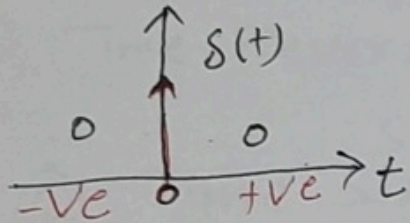


Digital Electronics input signals

input s/g 0, 1, 0, 1

Standard Signals

① Impulse signal



$$\delta(t) = 0 \quad t \neq 0$$

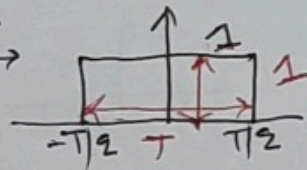
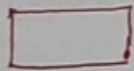
$$\neq 0 \quad t = 0$$

Area under impulse

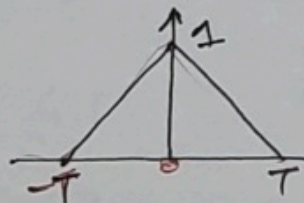
$$\int_{-\infty}^{\infty} \delta(t) \cdot dt = 1 \quad \text{Unit impulse}$$

$$\delta(t) \xrightarrow{\text{L.T}} 1$$

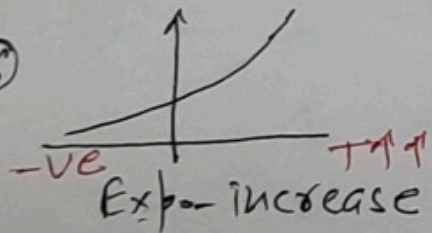
④ Rectangular Pulse



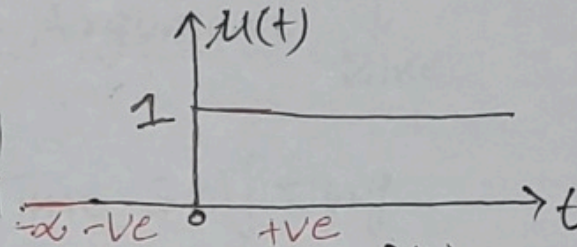
⑤ Triangular Pulse



⑥

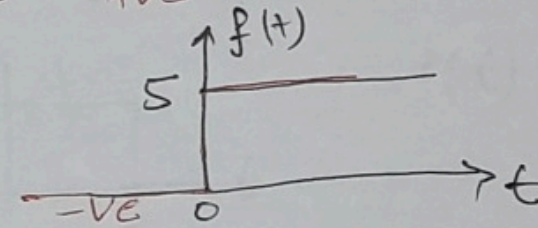


② Unit step signal



$$u(t) = 1 \quad t > 0$$

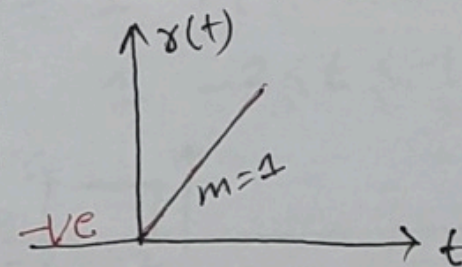
$$= 0 \quad t < 0$$



$$f(t) = 5u(t)$$

$$u(t) \xrightarrow{\text{L.T}} \frac{1}{s}$$

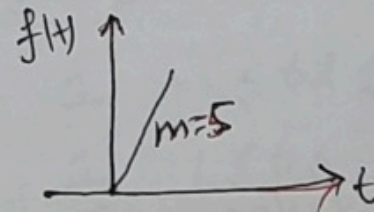
③ Ramp Signal



$$x(t) = t \quad t > 0$$

$$= 0 \quad t < 0$$

$$x(t) = t u(t)$$



$$f(t) = 5x(t)$$

$$= 5t u(t)$$

$$x(t) \xrightarrow{\text{L.T}} \frac{1}{s^2}$$

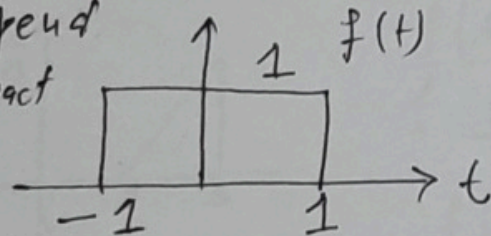
operation on signals

- ① Time scaling
- ② Time shifting
- ③ Time Reversal

Key Rule $\rightarrow$ whatever be limits
 $\downarrow$
 but Argument $\downarrow$
 same

① Time scaling

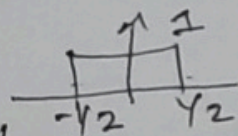
$a < 1$ expand
 $a > 1$ contract



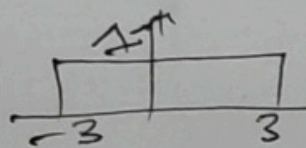
$$f(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(2t) = \begin{cases} 1 & -1 \leq 2t \leq 1 \\ 0 & \text{other} \end{cases}$$

$$= \begin{cases} 1 & -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 0 & \text{other} \end{cases}$$



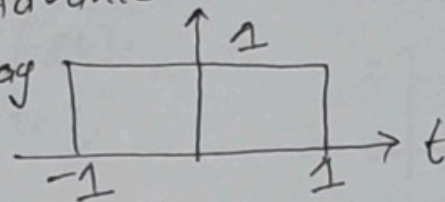
$$f\left(\frac{t}{3}\right) = \begin{cases} 1 & -1 \leq \frac{t}{3} \leq 1 \\ 1 & -3 \leq t \leq 3 \\ 0 & \text{other} \end{cases}$$



② Time shifting

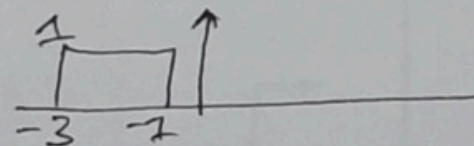
$t + T_0 \rightarrow$ Advance

$t - T_0 \rightarrow$ Delay

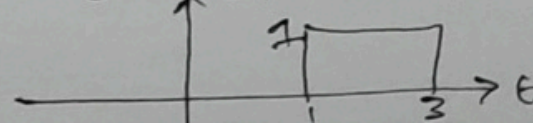


$$f(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

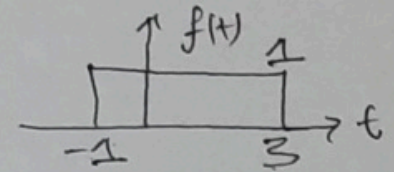
$$f(t+2) = \begin{cases} 1 & -1 \leq t+2 \leq 1 \\ 1 & -3 \leq t \leq -1 \end{cases}$$



$$f(t-2) = \begin{cases} 1 & -1 \leq t-2 \leq 1 \\ 1 & 1 \leq t \leq 3 \\ 0 & \text{other} \end{cases}$$



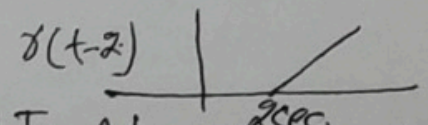
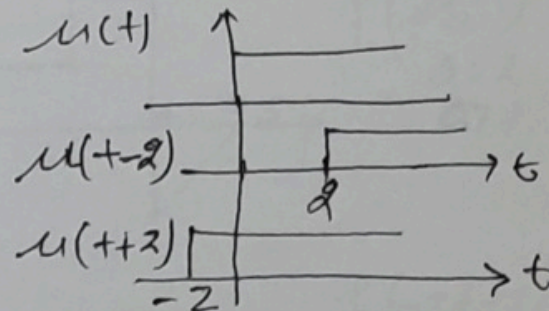
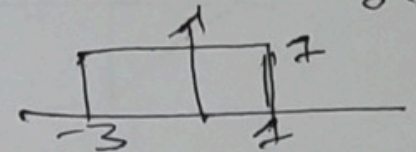
③ Time Reversal



$$f(t) = \begin{cases} 1 & -1 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$f(-t) = \begin{cases} 1 & -1 \leq (-t) \leq 3 \\ 0 & \text{other} \end{cases}$$

$$= \begin{cases} 1 & 1 \geq t \geq -3 \\ 0 & \text{other} \end{cases}$$



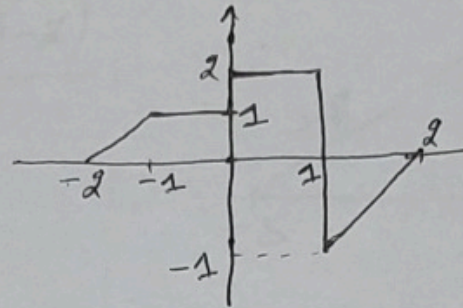
$t + T_0$ Adv
 $t - T_0$ Delay.

Problem on operation of signals

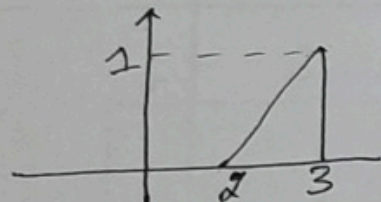
Always Remember this sq

- ① Time shifting
- ② Time scaling
- ③ Time Reversal.

② $f(t)$ is shown



① $f(t)$ is shown

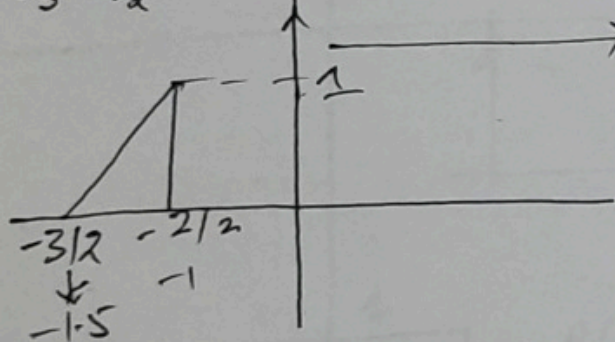
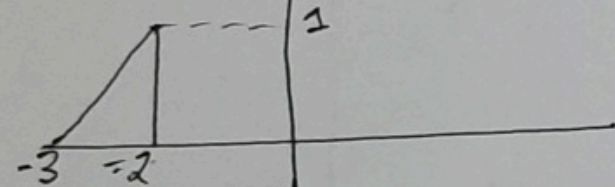


find

$$f(2t+5)$$

+ Draw it

$$f(t+5)$$

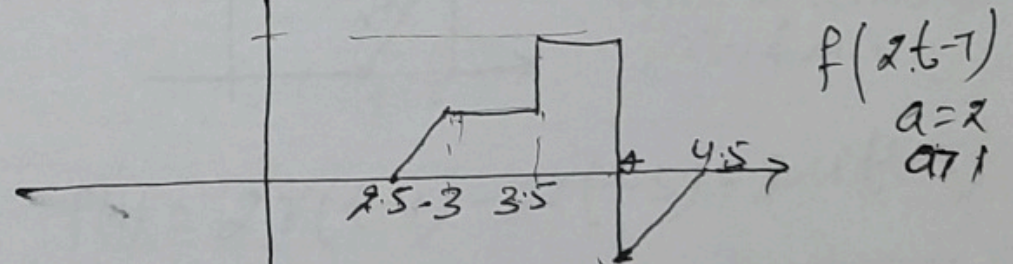
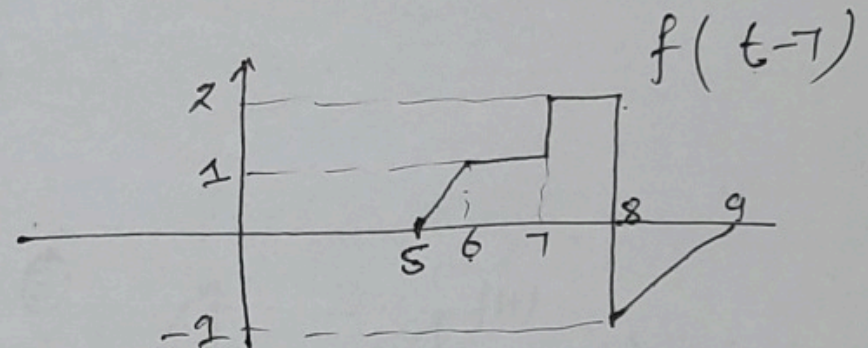


$$f(2t+5)$$

find

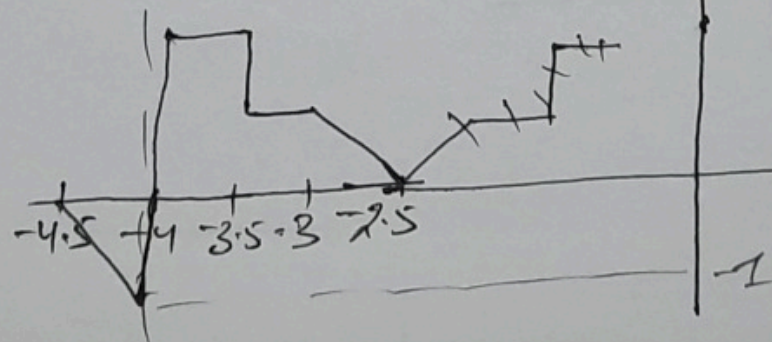
$$f(-2t-7)$$

$$f(-2t-7)$$



$$f(2t-7)$$

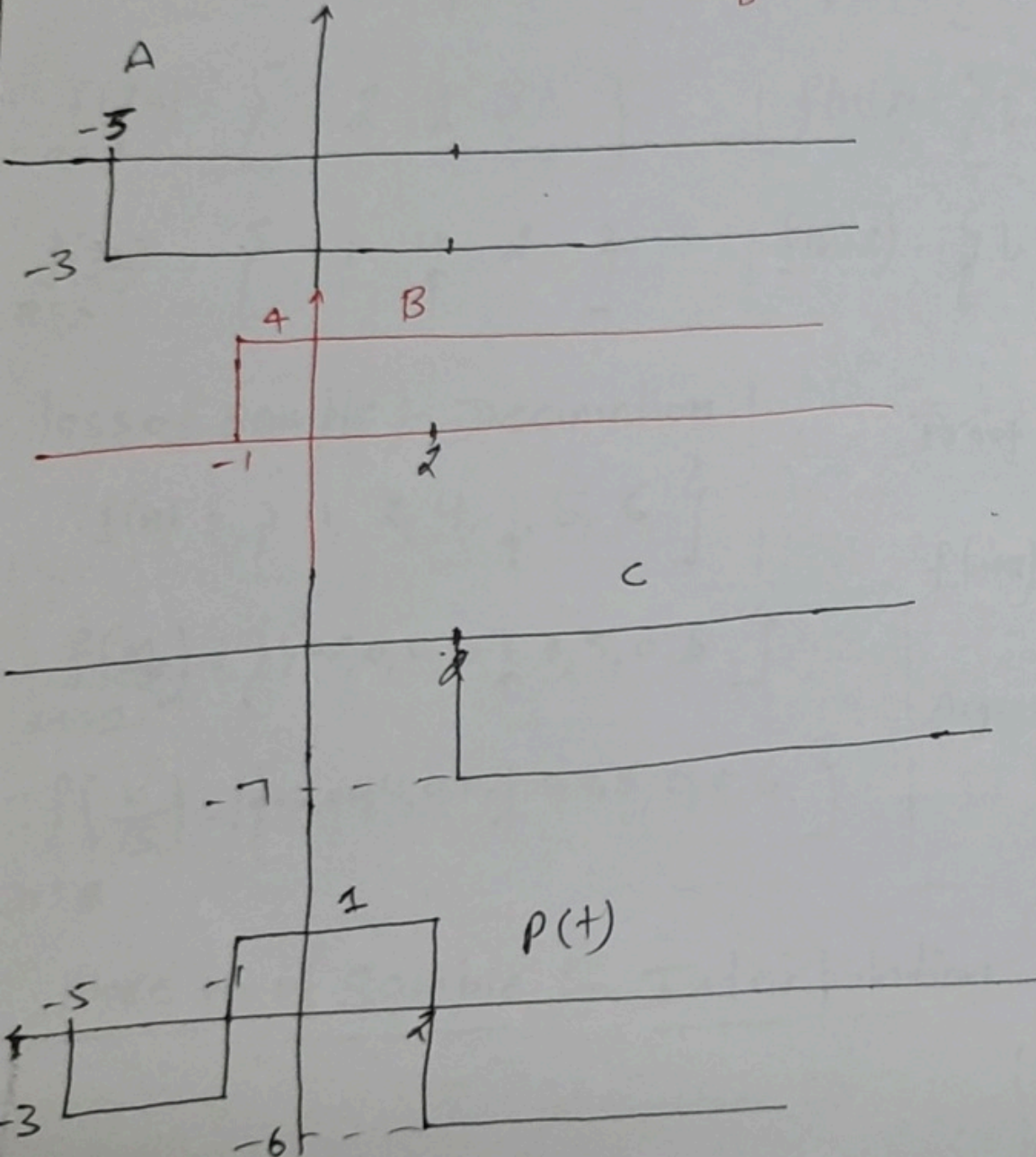
 $a=2$
 $a=7$



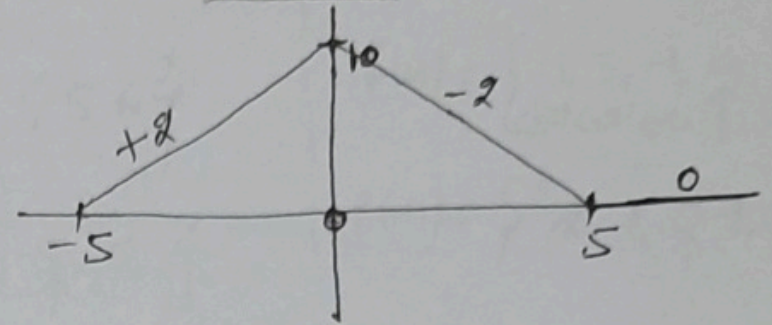
$$f(-2t-7)$$

① sketch the graph

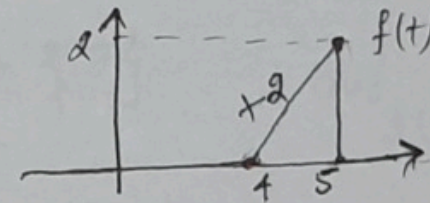
$$p(t) = \underbrace{-3u(t+5)}_A + \underbrace{4u(t+1)}_B - \underbrace{7u(t-2)}_C$$



② $f(t) = 2\delta(t+5) - 4\delta(t) + 2\delta(t-5)$



③



Write in terms of $\delta(t)$ & $u(t)$

$$f(t) = 2\delta(t-4) - 2\delta(t-5) - 2u(t-5)$$

Operation of Discrete Signal

Time Scaling

$$f(n) = \{1, 2, 3, \underset{\uparrow}{4}, 1, 3, 2\}$$

$$f(2n) = \{ \quad 2 \quad \underset{\uparrow}{4} \quad 3 \quad \}$$

$$f(3n) = \{ \quad 1 \quad \underset{\uparrow}{4} \quad 2 \quad \}$$

loss of sample :- Decimation

$$f(n) = \{1, 2, 4, \underset{\uparrow}{1}, 5, 6\}$$

$$f\left(\frac{n}{2}\right) = \{1, 0, 2, 0, 4, 0, \underset{\uparrow}{1}, 0, 5, 0, 6\}$$

$$f\left(\frac{n}{3}\right) = \{1, 0, 2, 0, 0, 4, 0, 0, \underset{\uparrow}{1}, 0, 0, 5, 0, 0, 6\}$$

more no of sample :- Interpolation

Time shifting

$$f(n) = \{1, 3, 2, \underset{\uparrow}{0}, 1, 5, 6\}$$

$$f(n+2) = \{1, 3, 2, 0, \underset{\uparrow}{1}, 5, 6\}$$

$$f(n-2) = \{1, \underset{\uparrow}{3}, 2, 0, 1, 5, 6\}$$

Proof $f(n) = \{1, \underset{\uparrow}{2}, 3\}$

$$f(n+2) = ?$$

Argument

$$f(n) = \begin{matrix} 1 & n=-1 \\ 2 & n=0 \\ 3 & n=1 \end{matrix}$$

$$\begin{matrix} 1 & n+2=-1 \Rightarrow n=-3 \\ 2 & n+2=0 \Rightarrow n=-2 \\ 3 & n+2=1 \Rightarrow n=-1 \end{matrix}$$

$$f(n+2) = \{1, 2, 3, \underset{\uparrow}{0}, 0\}$$

Time Reversal

$$f(n) = \{1, 3, -1, \underset{\uparrow}{0}, 1, 2\}$$

$$f(-n) = \{2, 1, \underset{\uparrow}{0}, -1, 3, 1\}$$

Proof $f(n) = \{1, \underset{\uparrow}{1}, 2\}$

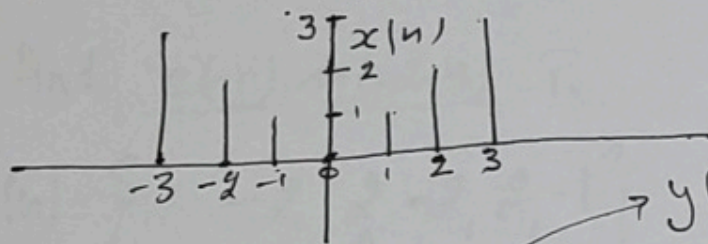
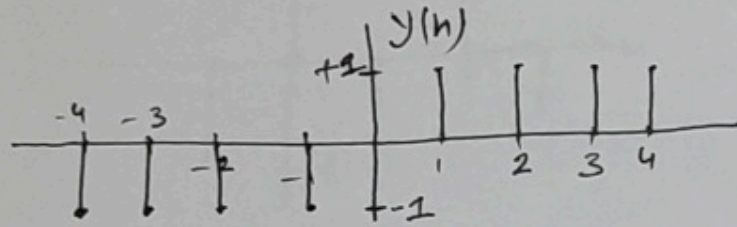
$$f(n) = \begin{matrix} 1 & n=-1, 0, 1 \end{matrix}$$

$$f(2n) = \begin{matrix} 1 & 2n=-1, 0, 1 \end{matrix}$$

$$\underline{1} \quad \left(n = -\frac{1}{2}, 0, \frac{1}{2} \right)$$

$$f(2n) = \{ \underset{\uparrow}{1} \}$$

Problem on operation of Discrete Signal.



find $x(2n)$, $y(1-n)$, $x(n-2) + y(n+2)$

$$x(n) = \{ 3, 2, 1, 0, 1, 2, 3 \}$$

$$x(2n) = \{ 3, 2, 0, 2 \}$$

$$y(n) = \{ -1, -1, -1, -1, 0, 1, 1, 1, 1 \}$$

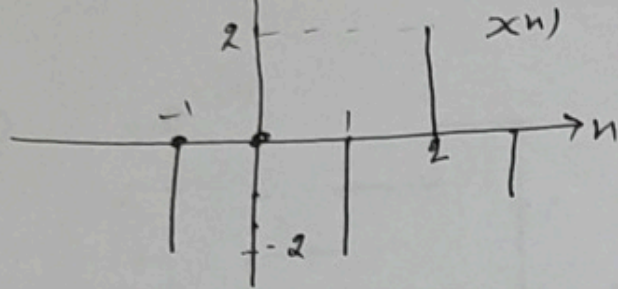
$$y(n+1) = \{ -1, -1, -1, -1, 0, 1, 1, 1, 1 \}$$

$$y(1-n) = \{ 1, 1, 1, 1, 0, -1, -1, -1, -1 \}$$

$$x(n-2) = \{ 3, 2, 1, 0, 1, 2, 3 \}$$

$$y(n+2) = \{ -1, -1, -1, -1, 0, 1, 1, 1, 1 \}$$

$$\{ -1, -1, -1, -1, 0, 4, 3, 2, 1, 1, 2, 3 \}$$



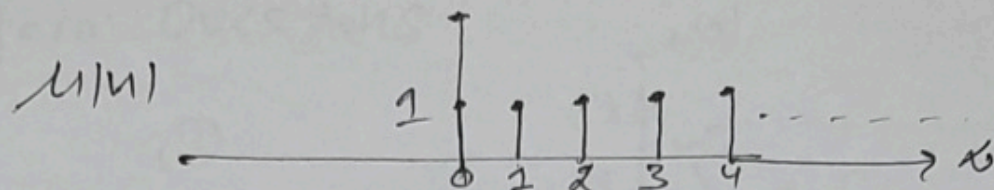
find $x(-n) \cdot u(-2-n)$ P.

$$x[n] = \begin{cases} -2, & \uparrow \\ 2, & \\ -2, & \\ 2, & \\ -1, & \end{cases}$$

① Time shifting

② " scaling

④ " Reversal



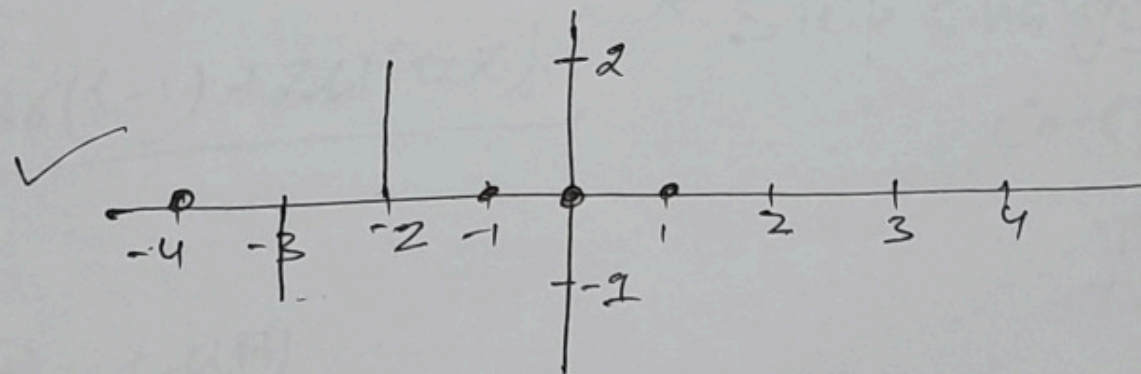
$$u[n] = \{0, 0, 0, \uparrow 1, 1, 1, 1, 1, \dots\}$$

$$u[n-2] = \{0, 0, \uparrow 0, 1, 1, 1, \dots\}$$

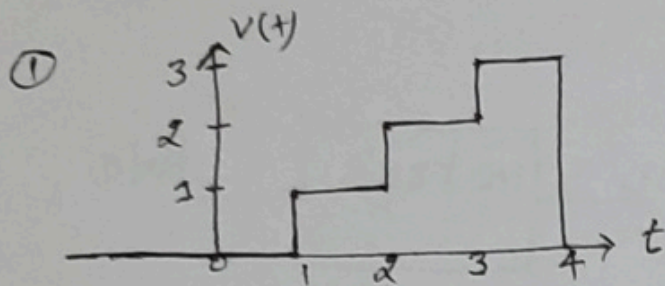
$$u[-n-2] = \{\dots, 1, 1, 1, 0, \uparrow 0, 0\}$$

$$x[-n] = \{\dots, -1, 2, -2, \uparrow 2, -2, \dots\}$$

$$x[-n] \cdot u[-n-2] = \{0, -1, 2, 0, \uparrow 0, 0\}$$



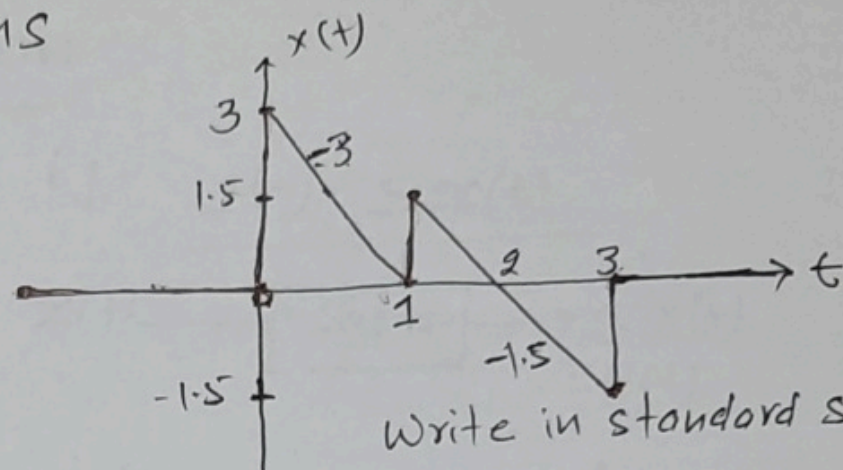
Waveform Questions



Write $v(t)$ in standard signal.

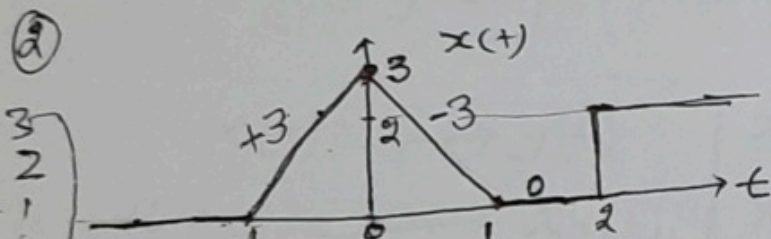
$$v(t) = 1u(t-1) + 1u(t-2) + 1u(t-3) - 3u(t-4)$$

③



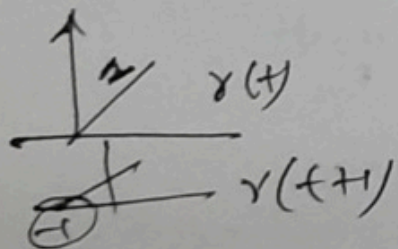
Write in standard signal

$$3u(t) - 3r(t) + 3r(t-1) + 1.5u(t-1) - 1.5r(t-1) + 1.5r(t-3) + 1.5u(t-3)$$



$x(t)$ in standard signal.

$$3r(t-1) - 6r(t) + 3r(t-1) + 2u(t-2)$$



$$\begin{aligned} r(t) &= t u(t) \\ 3r(t-1) &= 3(t-1)u(t-1) \\ 3r(t+1) &= 3(t+1)u(t+1) \end{aligned}$$

* slope change $\rightarrow$ Ramp
Co-eff $r(t-T)$

* step change $\rightarrow$ step s/g
Co-eff $u(t-T)$

$$\begin{aligned} T &\rightarrow -1 \text{ sec} \\ T &\rightarrow 1 \text{ sec} \end{aligned}$$